

30-Day DSA Roadmap for Placements

Week 1 - Master the Basics

Focus: Arrays, Strings, Sorting, Searching

Topics:

- Time & Space Complexity (Big O)
- Arrays (Basic + 2-pointer approach)
- Strings (Reversal, Palindrome, Frequency count)
- Sorting Algorithms: Bubble, Selection, Insertion, Merge, Quick
- Binary Search & its variations

Practice on: LeetCode Easy, GFG Arrays/Strings

Week 2 - Intermediate

Focus: Hashing, Recursion, Sliding Window, Stack & Queue

Topics:

- HashMap, HashSet usage
- Sliding Window Technique
- Recursion (Factorial, Fibonacci, Backtracking intro)
- Stack (Next Greater Element, Valid Parentheses)
- Queue (Circular Queue, Implement Queue using Stack)

Practice on: Sliding Window problems, GFG Stacks/Queues

Week 3 - Advanced

Focus: Linked List, Trees & Binary Trees

Topics:

- Singly & Doubly Linked List
- Fast and Slow pointers
- Binary Trees (Traversals: Inorder, Preorder, Postorder)
- BST (Search, Insert, Delete)
- Height, Diameter of Tree

Practice on: LeetCode Linked List, GFG Binary Tree

Week 4 - Final Topics + Revision

Focus: Graphs, Greedy, DP + Final Revision

Topics:

- Graphs (DFS, BFS, Topo Sort)
- Greedy (Activity selection, Fractional Knapsack)
- Dynamic Programming (Fibonacci, 0/1 Knapsack, LIS)
- Heap, Trie (only basic understanding)

Practice on: GFG Graphs, LeetCode DP

Daily Plan (4-6 hrs/day)

- 1 hr: Learn Concept (from GeeksforGeeks or YouTube)
- 3 hrs: Solve 4-5 problems (Easy to Medium)
- 1 hr: Revise yesterday's topic / Make notes
- 1 hr: Mock Test (on weekends)

Best Resources

- YouTube: CodeWithHarry, Apna College, Take U Forward
- Practice: LeetCode, GFG, InterviewBit
- Notes: Striver's DSA Sheet (takeuforward.org)

Tips

- Maintain a DSA Journal
- Join a discussion group
- Start mock interviews after 30 days